$\{4, 1, 6\}$

$6 \longrightarrow 4+1+6 = 11$

$\{4, 1\}$

$4 \longrightarrow 4+1 = 5$

$\{1\}$

$1 \longrightarrow 1 = 1$

$\overline{\quad 17 \quad}$

$\{3, 5, 1, -3\}$

$5 \longrightarrow 3+5+1+(-3) = 6$

$\{3, 1, -3\}$

$3 \longrightarrow 3+1+(-3) = 1$

$\{1, -3\}$

$1 \longrightarrow 1+(-3) = -2$

$\{-3\}$

$-3 \longrightarrow -3 = -3$

$\overline{\quad 2 \quad}$

[Break till 10:09 PM]

Selection Sort

A[ ] → 5 6 4 2

1st iter ⟶ min elem is swapped with A[0]

5 6 4 2 ⟶ $\boxed{2}$ 6 4 5

2nd iter ⟶ Next min elem is swapped with A[1]

2) 6 4 5   2 $\boxed{4}$ 6 5

sorted   rem

3rd iter ⟶ 2 4 (6 5)   2 4 $\boxed{5}$ 6

A ⟶ { 3 6 1 5 2 4 }

Iter 1 ⟶ 1 | 6 3 5 2 4

Iter 2 ⟶ 1 2 | 3 5 6 4

min → 3

Iter 3 ⟶ 1 2 3 | 5 6 4

Iter 4 ⟶ 1 2 3 4 | 6 5

Iter 5 ⟶ 1 2 3 4 5 6

```
for i -> 0 to n-2 {
    mn = i
    for j -> i to n-1 {
        if A[j] < A[mn] {
            mn = j
        }
    }
    tmp = A[mn]
    A[mn] = A[i]
    A[i] = tmp
}
```

$O(n^2)$ T.C always.

$O(1)$ S.C

$O(n)$ swaps

$O(n^2)$ comparisons.

After i iterations of selection sort,

$$A[0 \ldots i-1] = \text{sorted\_}A[0 \ldots i-1]$$

Selection sort is NOT stable sort.

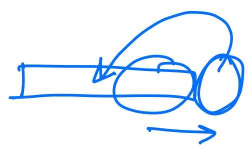
Insertion Sort

3 5 8 4 . . . . . .

3 4 5 8 9 6 . . . .

3 4 5 6 8 9 . . .

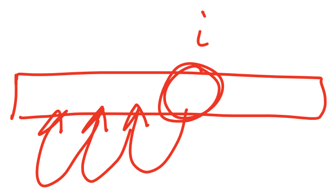
eg:- 6 8 2 7



6 8 2 7

6 2 8 7

2 6 8 7



2 6 7 8 → sorted.

2 6 7 8 3

3 8

3 7 8

2 3 6 7 8

arr → 3 4 5 6 7 1

1 3 4 5 6 7

```
for i → 1 to n-1 {
    for j → i-1 to 0 {
        if A[j+1] < A[j] {
            tmp = A[j]
            A[j] = A[j+1]
            A[j+1] = tmp
        }
        else {
            break
        }
    }
}
```

i=1 i=2

3 6 4 5

i=1 j=0

A[0+1] A[0]

6 > 3

i=2 j=1

A[1+1] A[1]

4 < 6

3 4 6 5

j=0

A[0+1] $\not<$ A[1]

i=3 j=2

3 $\boxed{4 \quad 5}$ 6

j=1

Worst case . $O(n^2)$ T.C

Best case → $O(n)$ T.C

Average case → $O(n^2)$ T.C

$O(1)$ S.C.

Worst case

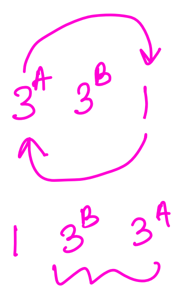
$O(n^2)$ comparisons

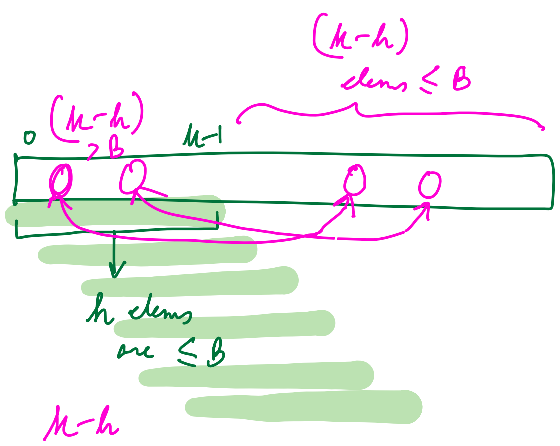
$O(n^2)$ swaps

After i iterations of insertion sort,

$A[0..i]$ is sorted BUT may not contain the minimum $(i+1)$ elements of the array.

Insertion sort is stable.





$\leq B \Rightarrow k$ elements